Final Exam (Spring 2025)

ELEC70082 – Distributed Optimisation and Learning

Instructor: Stefan Vlaski

28/04/2024, 10:00 – 13:00

Read all of the following information before starting the exam:

- The exam is open book. Feel free to use any written or printed material you have brought. You can use and refer to any results from the lecture notes that might be useful in answering the questions.
- Calculators or other electronic devices are not allowed for this exam.
- There is a total of 3 problems that total to 100 marks. There is a total of 4 pages.
- You have 3 hours (10:00 - 13:00) to complete the exam.

Question 1. The following questions require true/false answers and justification. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) (true/false) Incremental variants of decentralised algorithms generally exhibit improved performance in terms of their steady-state mean-squared deviation. [5 marks]

Solution. The statement is true. [2 marks]

The additional averaging arising from incremental steps reduces the impact of noise on the system. This is also evident from the performance expressions derived in lecture notes. [3 marks]

(b) (true/false) The bias of penalty-based decentralised optimisation algorithms (such as distributed gradient descent and diffusion) is amplified over sparse network topologies. [5 marks]

Solution. The statement is true. [2 marks]

Sparser topologies tend to exhibit λ_2 closer to 1, and the bias of decentralised algorithms is proportional to $\frac{1}{1-\lambda_2}$, which grows as $\lambda_2 \to 1$. [3 marks]

(c) (true/false) Suppose we employ exact gradients, and hence the gradient noise is zero. Then, for small step-sizes μ, the difference in steady-state mean-squared deviation between diffusion and Exact diffusion will be negligible. [5 marks]

Solution. The statement is false. [2 marks]

In the absence of gradient noise, primal-dual algorithms such as Exact diffusion converge exactly, while penalty-based algorithms such as diffusion exhibit a bias which is $O(\mu^2)$. [3 marks]

(d) (true/false) Decentralised algorithms for optimisation and learning offer privacy guarantees by design, because they avoid sharing raw data and instead share models. [5 marks]

Solution. The statement is false. [2 marks]

We saw in the chapter on differential privacy that private information can leak through model exchanges. Privacy can be ensured in rigorous ways through techniques like differential privacy, but this requires additional steps in the design of the algorithm. [3 marks]

(e) (true/false) Consider a collection of $K = 100$ agents. We examine two separate modes of operation. In arrangement (1), the agents are all connected to a fusion-center, and collaborate via centralised stochastic gradient descent with full participation. In arrangement (2), the agents are connected with each other over a sparse, but connected network topology and collaborate via the EXTRA algorithm. The stochastic gradient approximations and all other parameters are identical between the two arrangements. Then, for moderateley large step-sizes (small enough so the algorithms converge, but not so small that higher-order terms can be neglected) and gradient noise levels the two strategies will exhibit approximately the same steady-state mean-squared deviation. [5 marks]

Solution. The statement is false. [2 marks]

To first order, both strategies will have the same performance, but decentralised algorithms exhibit higher-order error terms that are not negligible for moderately large step-sizes. [3 marks]

(f) (true/false) Given two primitive matrices A_1 and A_2, we define the matrix: [5 marks]

$$A \triangleq \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Then A is also primitive.

Solution. The statement is false. [2 marks]

The matrix is block diagonal, so the powers of the matrix will never have non-zero off-diagonal entries, which is necessary for a primitive matrix. [3 marks]

(g) (true/false) The rate of convergence of the diffusion algorithm is unaffected by the mixing matrix of the combination matrix, provided that the step-size is sufficiently small. [5 marks]

Solution. The statement is true. [2 marks]

The rate of convergence is given by $\max\{1-\mu\nu+O(\mu^2), \lambda_2+O(\mu)\}$, which for small step-sizes is dominated by $1-\mu\nu+O(\mu^2)$. [3 marks]

(h) (true/false) Consider a star-graph, where agent 1 is connected to every other agent (and itself), and every other agent is only connected to agent one (and itself). Then, the entry of the Perron vector corresponding to agent 1 will be larger than all other entries. [5 marks]

Solution. The statement is true. [2 marks]

The Perron vector quantifies the centrality of a node in a network, and the central node in a star-graph is much more central than any other nodes. This can be established numerically as well, but the centrality argument is sufficient justification. [3 marks]

Question 2. We consider a collection of K agents, where each agent k is equipped with a local mean square error risk:

$$J_k(w) = \frac{1}{2}\mathbb{E}\Big(\boldsymbol{\gamma}_k - \boldsymbol{h}_k^{\mathsf{T}} w\Big)^2$$

The data follows the local linear models:

$$\boldsymbol{\gamma}_k = \boldsymbol{h}_k^{\mathsf{T}} w^o + \boldsymbol{v}_k$$

Note that the regressors $\boldsymbol{\gamma}_k$ and noise $\boldsymbol{v}_k$ are distinct across agents, but the linear model w^o is identical. We are interested in solving the consensus optimization problem:

$$w^o = \arg\min_{w\in\mathbb{R}^M} \frac{1}{K}\sum_{k=1}^{K} J_k(w)$$

(a) Suppose each agent k observes a single pair $\{\boldsymbol{h}_{k,i}, \boldsymbol{\gamma}_{k,i}\}$ at time i. All observations (across space and time) are independent. Formulate the recursions for the diffusion and for the NEXT algorithm when agents perform local elementary gradient approximations. [10 marks]

Solution. The diffusion recursions for linear regression with mini-batch stochastic gradients amount to: [5 marks]

$$\begin{aligned}
\boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1} + \mu \boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right) \\
\boldsymbol{w}_{k,i} &= \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i}
\end{aligned}$$

NEXT resembles the diffusion recursion, but uses a gradient-tracking variable instead of the local gradient approximations: [5 marks]

$$\begin{aligned}
\boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1} - \mu \boldsymbol{g}_{k,i-1} \\
\boldsymbol{w}_{k,i} &= \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i} \\
\boldsymbol{g}_{k,i} &= \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{g}_{\ell,i-1} - \boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i}\right) + \boldsymbol{h}_{k,i-1}\left(\boldsymbol{\gamma}_{k,i-1} - \boldsymbol{h}_{k,i-1}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right)
\end{aligned}$$

(b) Suppose agents only observe a pair $\{\boldsymbol{h}_{k,i}, \boldsymbol{\gamma}_{k,i}\}$ with probability π_k, and no data otherwise. How would you need to adjust the recursions from part (a) to ensure that you continue to have a working algorithm with unbiased local gradient approximations? [10 marks]

Solution. Similar to asynchronous implementations we have seen in lectures and problem sets, agents will need to scale their local updates to ensure the effective gradient approximation is unbiased (see for example Eq. (7.57) in the notes). This means that locally, agents need to construct:

$$\widehat{\nabla J}^{\,\text{diff}}(\boldsymbol{w}_{k,i-1}) \triangleq \begin{cases} -\frac{1}{\pi_k}\boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right), & \text{with probability } \pi_k \\ 0, & \text{otherwise.} \end{cases}$$ [4 marks]

or

$$\widehat{\nabla J}^{\,\text{NEXT}}(\boldsymbol{w}_{k,i}) \triangleq \begin{cases} -\frac{1}{\pi_k}\boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i}\right), & \text{with probability } \pi_k \\ 0, & \text{otherwise.} \end{cases}$$

Note that the time-indexing convention here follows the one used in the lecture notes (see, e.g., Section 16.4) with differences between the diffusion and NEXT approximations, but this is largely irrelevant for the performance of the algorithm. Other variants are possible and will receive full credit if they yield a working algorithm. With these constructions, the recursion for diffusion becomes:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} - \mu\widehat{\nabla J}^{\text{diff}}(\boldsymbol{w}_{k,i-1}) = \begin{cases} \boldsymbol{w}_{k,i-1} + \mu\boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right), & \text{w.p. } \pi_k, \\ \boldsymbol{w}_{k,i-1}, & \text{otherwise.} \end{cases} \quad \text{[3 marks]}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i}$$

And for NEXT:

$$\begin{aligned} \boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1} - \mu\boldsymbol{g}_{k,i-1} \quad \text{[3 marks]} \\ \boldsymbol{w}_{k,i} &= \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i} \\ \boldsymbol{g}_{k,i} &= \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{g}_{\ell,i-1} + \widehat{\nabla J}^{\text{NEXT}}(\boldsymbol{w}_{k,i}) - \widehat{\nabla J}^{\text{NEXT}}(\boldsymbol{w}_{k,i-1}) \end{aligned}$$

(c) Determine expressions for the steady-state mean-squared deviation of both the diffusion and NEXT implementations from part (b). [10 marks]

Solution. The performance expressions can be inferred from Table 16.2 in the notes:

$$\limsup_{i\to\infty} \mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma^2}{\nu}\right) + O\left(\frac{\mu^2\lambda_2^2 K\sigma^2}{1-\lambda_2}\right) + O\left(\frac{\mu^2\lambda_2^2\sum_{k=1}^K \|\nabla J_k(w^o)\|^2}{K(1-\lambda_2)^2}\right) \quad \text{[2 marks]}$$

Because the local models w^o are identical, hence w_k^o and $\nabla J_k(w^o) = 0$, and the bias term vanishes:

$$\limsup_{i\to\infty} \mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma^2}{\nu}\right) + O\left(\frac{\mu^2\lambda_2^2 K\sigma^2}{1-\lambda_2}\right) \quad \text{[2 marks]}$$

For the NEXT algorithm, we find from the same table:

$$\limsup_{i\to\infty} \mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma^2}{\nu}\right) + O\left(\frac{\mu^2 K\sigma^2}{1-\lambda_2}\right) \quad \text{[3 marks]}$$

The bounds are very similar, indeed the bound for the diffusion algorithm is smaller than that for the NEXT algorithm, because the local objectives share a minimizer. The only missing quantity is the aggregate gradient noise σ^2, for which we find from Eqs. (7.45), (7.60) and (15.44):

$$\sigma^2 = \frac{1}{K^2}\sum_{k=1}^K \frac{\sigma_k^2}{\pi_k} = \frac{1}{K^2}\sum_{k=1}^K \frac{\sigma_{v_k}^2 \operatorname{Tr}(R_{h_k})}{\pi_k} \quad \text{[3 marks]}$$

Plugging this relation into both performance bounds gives the result.

Question 3. We consider an incremental variant of the exact consensus algorithm for tracking the average of a collection of signals $\{\boldsymbol{g}_{k,i}\}_{k=1}^{K}$ over a network:

$$\begin{aligned}\boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1} + \boldsymbol{g}_{k,i} - \boldsymbol{g}_{k,i-1} \\ \boldsymbol{w}_{k,i} &= \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{\psi}_{\ell,i}\end{aligned}$$

The signals $\boldsymbol{g}_{k,i} \in \mathbb{R}$ are scalars, and we initialize $\boldsymbol{w}_{k,0} = \boldsymbol{g}_{k,0}$. The combination weights are collected in a primitive matrix A that satisfies:

$$\begin{aligned}A &= A^{\mathsf{T}} \\ A\mathbb{1} &= \mathbb{1}\end{aligned}$$

We model the signals as follows:

$$\boldsymbol{g}_{k,i} = \begin{cases}\boldsymbol{g}_{k,i-1}, & \text{with probability } 1-\epsilon, \\ \boldsymbol{g}_{k,i-1} + \boldsymbol{r}_{k,i}, & \text{otherwise.}\end{cases}$$

where $\boldsymbol{r}_{k,i} \sim \mathcal{N}(0, \sigma_r^2)$. This means that at any time i, the signal $\boldsymbol{g}_{k,i}$ will remain at $\boldsymbol{g}_{k,i-1}$ with probability $1-\epsilon$. With probability ϵ, it will move by a random value $\boldsymbol{r}_{k,i}$, which is modelled as a normally distributed random variable with common variance σ_r^2. This means that ϵ close to 0 indicates a stable environment, while ϵ close to 1 indicates a highly volatile environment.

(a) Determine an expression for the mean and variance of $\boldsymbol{g}_{k,i} - \boldsymbol{g}_{k,i-1}$. [10 marks]

Solution. We have:

$$\boldsymbol{v}_{k,i} = \boldsymbol{g}_{k,i} - \boldsymbol{g}_{k,i-1} = \begin{cases}0, & \text{with probability } 1-\epsilon, \\ \boldsymbol{r}_{k,i}, & \text{otherwise.}\end{cases} \quad \text{[3 marks]}$$

Hence:

$$\mathbb{E}\boldsymbol{v}_{k,i} = \mathbb{E}\,\{0\} \times (1-\epsilon) + \mathbb{E}\boldsymbol{r}_{k,i} \times \epsilon = 0 \quad \text{[3 marks]}$$

For the variance:

$$\mathbb{E}\boldsymbol{v}_{k,i}^2 = \mathbb{E}\left\{0^2\right\} \times (1-\epsilon) + \mathbb{E}\boldsymbol{r}_{k,i}^2 \times \epsilon = \sigma_r^2 \epsilon \quad \text{[4 marks]}$$

(b) Determine an expression for the network centroid: [10 marks]

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{w}_{k,i}$$

in terms of the signals $\boldsymbol{g}_{k,i}$.

Solution. Since A is doubly-stochastic, it holds that:

$$\begin{aligned}\boldsymbol{w}_{c,i} &= \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{w}_{k,i} = \frac{1}{K}\sum_{k=1}^{K}\sum_{\ell \in \mathcal{N}_k} a_{\ell k}\left(\boldsymbol{w}_{\ell,i-1} + \boldsymbol{g}_{\ell,i} - \boldsymbol{g}_{\ell,i-1}\right) \quad \text{[7 marks]} \\ &= \frac{1}{K}\sum_{\ell=1}^{K}\left(\sum_{k=1}^{K} a_{\ell k,i}\right)\left(\boldsymbol{w}_{\ell,i-1} + \boldsymbol{g}_{\ell,i} - \boldsymbol{g}_{\ell,i-1}\right)\end{aligned}$$

$$= \frac{1}{K}\sum_{\ell=1}^{K}\left(\boldsymbol{w}_{\ell,i-1} + \boldsymbol{g}_{\ell,i} - \boldsymbol{g}_{\ell,i-1}\right)$$
$$= \boldsymbol{w}_{c,i-1} + \boldsymbol{g}_{c,i} - \boldsymbol{g}_{c,i-1}$$

After iterating, we find: [3 marks]

$$\boldsymbol{w}_{c,i} = \boldsymbol{g}_{c,i}$$

(c) Determine an expression for the rate of convergence of the tracking error [10 marks]

$$\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2,$$

relative to the signal centroid $\boldsymbol{g}_{c,i} = \frac{1}{K}\sum_{k=1}^{K}\boldsymbol{g}_{k,i}$, as well as its limiting value:

$$\limsup_{i\to\infty}\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2$$

Solution. From our result in part (b) and the fact that A is doubly-stochastic, it holds that:

$$\sum_{k=1}^{K}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2 = \sum_{k=1}^{K}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 = \left\|\left(I - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\otimes I\right)\boldsymbol{w}_i\right\|^2 \quad \text{[2 marks]}$$

and hence

$$\left(I - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\otimes I\right)\boldsymbol{w}_i = \left(I - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\otimes I\right)\mathcal{A}^{\mathsf{T}}\left(\boldsymbol{w}_{i-1} + \boldsymbol{v}_i\right) \quad \text{[2 marks]}$$
$$= \left(\mathcal{A}^{\mathsf{T}} - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\otimes I\right)\left(I - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\right)\boldsymbol{w}_{i-1} + \left(\mathcal{A}^{\mathsf{T}} - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\otimes I\right)\boldsymbol{v}_i$$

Squaring, taking expectations, removing cross-terms and using the expression for the variance from part (b):

$$\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{g}_{c,i}\|^2 \le \lambda_2^2\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i-1} - \boldsymbol{g}_{c,i-1}\|^2 + \lambda_2^2\sum_{k=1}^{K}\sigma_r^2\epsilon \quad \text{[2 marks]}$$
$$= \lambda_2^2\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i-1} - \boldsymbol{g}_{c,i-1}\|^2 + \lambda_2^2 K\sigma_r^2$$

Hence, the rate of convergence is given by λ_2^2. Iterating, we can find: [2 marks]

$$\limsup_{i\to\infty}\sum_{k=1}^{K}\mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 \le \frac{\lambda_2^2 K\sigma_r^2}{1-\lambda_2^2} \quad \text{[2 marks]}$$